We know that 3 is a zero of the polynomial x^2-5x+6 and 3 is a root or solution of the equation $x^2-5x+6=0$. We note that $\cos x=\sin x$ and $\cos x+\sin x=1$ are also equations in one variable x. However, $\cos x-\sin x$ and $\cos x+\sin x-1$ are not polynomials and hence $\cos x=\sin x$ and $\cos x+\sin x=1$ are not "polynomial equations". We are going to consider only "polynomial equations" and equations which can be solved using polynomial equations in one variable.

We recall that $\sin^2 x+\cos^2 x=1$ is an identity on $\mathbb{R}$, while $\sin x+\cos x=1$ and $\sin^3 x+\cos^3 x=1$ are equations.

It is important to note that the coefficients of a polynomial can be real or complex numbers, but the exponents must be nonnegative integers. For instance, the expressions $3x^{-2}+1$ and $5x^{\frac{1}{2}}+1$ are not polynomials. We already learnt about polynomials and polynomial equations, particularly about quadratic equations. In this section let us quickly recall them and see some more concepts.

3.2.2 Quadratic Equations

For the quadratic equation $ax^2+bx+c=0$, b^2-4ac is called the discriminant and it is usually denoted by Δ. We know that $\frac{-b+\sqrt{\Delta}}{2a}$ and $\frac{-b-\sqrt{\Delta}}{2a}$ are roots of the $ax^2+bx+c=0$. The two roots together are usually written as $\frac{-b\pm\sqrt{b^2-4ac}}{2a}$. It is unnecessary to emphasize that $a\neq 0$, since by saying that ax^2+bx+c is a quadratic polynomial, it is implied that $a\neq 0$.

We also learnt that $\Delta=0$ if, and only if, the roots are equal. When a,b,c are real, we know

- $\Delta>0$ if, and only if, the roots are real and distinct
- $\Delta<0$ if, and only if, the quadratic equation has no real roots.

3.3 Vieta's Formulae and Formation of Polynomial Equations

Vieta's formulae relate the coefficients of a polynomial to sums and products of its roots. Vieta was a French mathematician whose work on polynomials paved the way for modern algebra.

3.3.1 Vieta's formula for Quadratic Equations

Let α and β be the roots of the quadratic equation $ax^2+bx+c=0$. Then

$$ax^2+bx+c=a(x-\alpha)(x-\beta)=ax^2-a(\alpha+\beta)x+a(\alpha\beta)=0.$$

Equating the coefficients of like powers, we see that

$$\alpha+\beta=\frac{-b}{a} \text{ and } \alpha\beta=\frac{c}{a}.$$

So **a** quadratic equation whose roots are α and β is $x^2-(\alpha+\beta)x+\alpha\beta=0$; that is, **a** quadratic equation with given roots is,

$$x^2-(\text{sum of the roots})\,x+\text{product of the roots}=0. \qquad \ldots (1)$$

Note

The indefinite article **a** is used in the above statement. In fact, if $P(x)=0$ is a quadratic equation whose roots are α and β, then $cP(x)$ is also a quadratic equation with roots α and β for any non-zero constant c.

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